Project handoff

Energy-aware learning to defer for sentiment analysis on battery-powered devices

Scope: everything done on the project since Review 2 (22 September 2026), up to 26 October 2026.

Purpose: give anyone picking the project up the full context: what was decided, what was built and measured, every number with its source, what it cost, what was corrected along the way, where every file is, and what is still open.

Project folder: /Volumes/Ehsan/AI Sentiment (external drive). Nothing in it is committed to git yet (see section 12).

How to read it: section 1 is a one-page summary. Sections 2–10 go through the work in order. Section 11 lists corrections, section 12 open items, and section 13 is the reproduce-and-file map.

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1. Summary

At Review 2 the plan was a single CPU model: TF-IDF + logistic regression with one extra VADER lexicon feature. That experiment was never run. The project instead became a two-tier system: a 256 KB hashed n-gram model on an ESP32-S3 microcontroller answers most reviews itself and sends only some over Wi-Fi to a transformer on a Raspberry Pi Zero 2 W hub. The research question became which reviews are worth sending: the device's own confidence (confidence routing), or a second small model trained to predict whether the hub would fix the answer (learned deferral).

Headline results

- **Device model:** 256 KB of int8 weights, 92.69% on the official IMDb test set (trained on 1.94M reviews; 92.32% after the stricter second leakage clean).
- **Hubs:** DistilBERT 93.51% on the official 20k training reviews; retrained with 1M extra leak-free reviews, DistilBERT 95.83% and RoBERTa-base 96.72%.
- **Main research result (negative):** learned deferral does not beat confidence routing. Across three hubs it stays within 0.14 accuracy points at every tested send rate, and none of 15 differences survives a Bonferroni correction. Reason: stronger hubs rarely break a correct device answer (1.2–1.5% vs 3.9%), which removes learned deferral's easy win.
- **Energy (modelled, not measured):** with the strong hubs, sending about 8% of reviews closes half the device–hub accuracy gap at about 12× less device energy than sending every review; sending 15–17% closes three quarters at 5.8–6.5× less.
- **Record attempt:** an ensemble of ELECTRA-large + DeBERTa-v3-large + the 1 MB n-gram model scores 97.40% on IMDb test (95% CI 97.21–97.58%), above the best peer-reviewed 97.1%, in the extra-training-data setting.
- **Cost:** $20.27 of Modal cloud GPU time, itemised per run, all paid from free credits ($0 billed).

Deliverables produced

- IEEE conference paper, 5 pages, Word and PDF (paper/).
- 22-slide presentation split across three presenters (Aditya slides 1–8, Danyson 9–15, Ehsan 16–22), with word-for-word speaker notes (slides/).
- Presenter scripts as Word, PDF, Markdown and a shared online Doc.
- A live laptop demo that runs the real models and shows routing and simulated energy per review (demo/).
- Updated ROADMAP.md (Phases 3–6 and a Phase 5 follow-up section).

Most important open items

- **Nothing is committed to git.** The repository has no commits; every file is untracked.
- **Hardware (Phase 7) is not started:** the energy numbers use datasheet currents with placeholder timings.
- Two small wording fixes are pending in the paper and slides, and the ROADMAP's Phase 6 energy figures should be aligned with the paper's (section 11).

2. Starting point: Review 2

Review 2 (deck review2_update.pptx, dated 22 Sep 2026; presented by Ehsan Abdulkhader, Danyson Antony and Aditya Achar) proposed:

- Keep the Review 1 baseline: TF-IDF unigrams + bigrams → logistic regression, CPU only.
- Add one feature: VADER's rule-based compound sentiment score, stacked onto the TF-IDF matrix (hybrid lexicon + ML).

- Rule out Jev / Open-Jev (GPU-scale models, unpublished method) as contrary to the CPU-only thesis.
- Results were placeholders ("TBD"); calibration was a stretch goal.

What happened to that plan: it was replaced before being run; there is no VADER code in the repository. The VADER paper is kept in papers/09_earlier_approach_vader for a "what we tried before" section. The new direction is the two-tier device + hub system in section 4.

A project-update deck (slides/project_update.pptx, 19 Oct) already reported the state as of that date. Everything after 19 Oct (the retrained hubs, the paper, the final talk, the demo and the corrections) is new in this handoff, and a few figures in that deck are now out of date (section 11).

Team: Ehsan Abdulkhader (24BCT0089), Danyson Antony (24BAI0196) and Aditya Achar (24BCI0250), the same team as Review 2. The paper, slides and scripts all list these three.

3. Timeline

Dates come from the job logs in outputs/logs/ (time the log was last written) and from Modal's billing report.

When	What was done
22 Sep	Review 2: TF-IDF + VADER plan presented.
19 Oct, early	Phase 0 baseline: TF-IDF + LR (01:27), ESP32 size study (01:43), DistilBERT fine-tune (02:20), confidence cascade (02:25).
19 Oct, morning	Phase 1 data: Kaggle IMDb and Amazon downloads and pool builds (03:24–04:49). Phase 4 hub labels (09:45). Phase 2 device model at 1.94M reviews (09:51). Phases 5–6 on this first-round data: learned deferral vs confidence routing (level, ±0.1 pts) and the energy model (09:56).
19 Oct, late morning	Leakage clean round 2 after an adversarial review: test movies, test authors, near-duplicates; splits_v2 (11:01).
19 Oct, afternoon	Record attempt on Modal H100s: RoBERTa-large official reproduction, then RoBERTa-large, ELECTRA-large and DeBERTa-v3-large with 1M extra reviews, plus long-review predictions (12:12–15:57). n-gram models retrained on v2 data (14:05). Stacked ensemble. Project-update deck (16:19).
25–26 Oct	Phase 5 follow-up: hubs retrained on 1M leak-free reviews (DistilBERT, RoBERTa-base), deferral slice re-drawn, routers retrained, cascade and energy rerun, significance tests added.
26 Oct	IEEE paper; presentation (12 then 22 slides); presenter scripts (Doc, Word, PDF); cost itemisation and correction; laptop demo; 64 KB → 128 KB router correction; this handoff.

4. The system

Device: an ESP32-S3 microcontroller on battery (512 KB on-chip SRAM, about $5). It hashes the review's words and word pairs into 2^18 buckets, adds up one int8 weight per bucket (divided by √n, n = distinct buckets) and applies a sigmoid to get p(x), the probability the review is positive. Model: 262,144 int8 weights = 256 KB, plus one scale (0.3075) and one bias (0.0152). No vocabulary is stored.

Router: a second model on the device computes g(x), the expected accuracy gain of sending the review. The device sends the review only if g(x) > λ, where λ is the price of one Wi-Fi send in accuracy. Sweeping λ traces accuracy against battery use.

Hub: a Raspberry Pi Zero 2 W (512 MB RAM, about $15) on the same local network, running an int8 transformer (DistilBERT or RoBERTa-base). It returns the label.

Two routing rules compared: confidence routing sends reviews with the highest 1 − max(p, 1 − p). Learned deferral is a 3-class logistic regression predicting gain = (hub right) − (device right) ∈ {−1, 0, +1} from 2^16 hashed n-gram buckets, the device's confidence (10 bins) and review length (9 bins); g(x) = P(+1) − P(−1).

Memory: the router needs two int8 weight tables of 65,536 entries (fix and break logits relative to no change) = 128 KB. Both device models together use 384 KB of the 512 KB SRAM.

Use cases (from the roadmap): social and assistive robots (motors are only 16.6–54.6% of measured robot power; Liu et al. 2025), off-grid messengers (Meshtastic on an ESP32-S3 LILYGO T-Deck), wearables with a phone as hub, smart toys and battery feedback kiosks.

5. Data and leakage control (Phase 1)

Labels: IMDb rating ≤ 4/10 negative, ≥ 7/10 positive (Maas et al. 2011 rule); Amazon 1–2 stars negative, 4–5 positive; middle and missing ratings dropped; Amazon titles dropped because they state the rating. **Official splits:** IMDb train 25,000 → 20,000 train + 5,000 validation (fixed seed); test 25,000, never changed, scored once per final system.

Round 1: build the pool and remove copies of the benchmark

Step	Kaggle IMDb	Amazon Movies & TV
Reviews read	5,571,499	17,328,314
No usable rating	1,384,838	1,260,217
Too short	614	3,353,755
Duplicates within the pool	39,243	762,073
Copies of official test reviews (exact / prefix / suffix)	23,636 (22,766 / 537 / 333)	182
Copies of validation reviews	4,695	44
Copies of training reviews	18,745	147
Kept	4,099,728 (1,135,184 neg / 2,964,544 pos)	11,951,896

22,766 of the 25,000 official test reviews (91%) were in the public scrape word for word. Training on them would have produced a fake, too-good score.

Round 2: after an adversarial review

Flag	Reviews
Reviews of official test-set movies (title + year match, roman-numeral suffixes handled)	163,409
Reviews by any of the 16,819 authors who also wrote a test review	635,853
Near-duplicates sharing ≥ 30% of a val/test review's word 8-grams	443
Eligible IMDb reviews left (movie- and author-disjoint from test)	3,336,925

Splits used

Split	Reviews	Use
v1 chip_train (round 1)	1,920,408	Device model v1 (with the 20k official reviews: 1.94M)
v1 defer (round 1)	300,000	Deferral labels for the first-round experiments (weak hub)

Split	Reviews	Use
v2 chip_train (round 2)	1,532,574	Device model v2; the first 1,000,000 rows train the 1M hubs and record models
v2 defer (round 2)	300,000	Deferral labels for the retrained hubs; verified 0 overlap with hub training
v2 val_pool	50,000	Held-out 5% of movies, length-matched to test; used to pick checkpoints and ensembles
Ablation sets	163,298 / 599,062 / 1,253,450	Test-movie reviews / test-author reviews / long Amazon reviews

Why the deferral slice was re-drawn: 94,194 of the 300,000 v1 deferral reviews (31%) are inside the first 1M rows of v2 chip_train, i.e. inside the retrained hubs' training data. Labels from it would have flattered the hub, so the 1M-hub experiments use the v2 slice (0 overlap, checked by hashing every review).

6. Models: baseline, device model, hubs

Phase 0: baseline cascade (official 20k training reviews)

System	Test accuracy	Sent to transformer	Reviews/s (M3 Pro)
TF-IDF + logistic regression (C by 3-fold CV)	90.28%	0%	6,822
DistilBERT (2 epochs, batch 16, lr 3e-5, head 128 + tail 382 tokens)	93.51%	100%	133
Confidence cascade, τ = 0.64 (fewest sends within 0.5 pts of DistilBERT on val)	92.50%	10.0%	960
Stacked ensemble of both (upper bound)	93.95%	100%	131

At least one of the two models is right on 96.46% of test reviews. On validation the cascade first matches DistilBERT at 14.5% sent; its most accurate validation point is 26% sent (the ROADMAP's “matches DistilBERT at 28% sent” comes from this curve). DistilBERT trained in 47 minutes on the laptop GPU.

ESP32 feasibility (20k training reviews, validation accuracy, int8)

Hash table	16 KB (2^14)	64 KB (2^16)	256 KB (2^18)	1 MB (2^20)
1–2-grams, Naive Bayes weighting	86.6%	89.3%	90.3%	90.6%

1–3-grams were no better. Naive Bayes weighting (Wang & Manning 2012) helps at small data and is folded into the weights at no device cost. 256 KB was chosen: nearly the 1 MB accuracy at a quarter of the memory.

Phase 2: device model trained on millions of reviews

Method: hashed unigrams + bigrams, binary and L2-normalised; Naive Bayes log-count ratios folded into the weights; logistic loss with averaged SGD, 5 passes in 100k-review chunks; regularisation chosen on validation; weights rounded to int8 with one float scale.

Training reviews	20k	100k	300k	1M	1.94M
256 KB validation accuracy	90.18%	91.10%	91.82%	92.30%	92.76%

- **v1 (round-1 data, 1.94M reviews):** 256 KB 92.69% test (92.76% val); Naive Bayes off 92.72% val; 1 MB 93.12% val.
- **Ablations:** adding reviews of test-set movies changes test accuracy by +0.02 points (leakage of that kind barely matters for this model); adding 1.25M long Amazon reviews changes it by −0.13 (Amazon was dropped).
- **v2 (round-2 data, 1.55M reviews):** 256 KB 92.32% test, 1 MB 92.80% test. Lower than v1 because stricter cleaning removed about 390k training reviews. v2 is the device model for all retrained-hub results and the demo.

Phase 3: hubs

Hub	Training reviews	Epochs	Batch	Learning rate	Hardware	Val	val_pool	Test
DistilBERT, 20k	20,000	2	16	3e-5	Laptop GPU, 47 min	93.16%	–	93.51%
DistilBERT, +1M	1,020,000	1	32	3e-5	H100, ≈2,240/s	95.68%	96.05%	95.83%
RoBERTa-base, +1M	1,020,000	1	32	2e-5	H100, ≈1,170/s	96.46%	96.80%	96.72%

Shared recipe for the +1M runs (src/train_hub.py via src/modal_app.py): 1,020,000 ÷ 32 = 31,875 optimiser steps; AdamW, warm-up over 6% of steps then linear decay; bfloat16; length-bucketed batches; long reviews keep the first 128 and last 382 tokens; accuracy on val_pool every 2,000 steps, best checkpoint kept. RoBERTa-base reused RoBERTa-large's cached tokens (identical tokenizer), saving a tokenize run. As int8 models they should need about 67 MB and 125 MB on the Pi; RoBERTa-base needs about twice DistilBERT's compute per review (not yet measured on the Pi).

7. Record attempt on IMDb

Goal: test how high accuracy can honestly go with leak-free extra data. All choices (model variants, whether to read long reviews in sliding windows, ensemble members, stacker) were made on val_pool; the test set was scored once for the final ensemble.

System	Learning rate	val_pool	Test
RoBERTa-large, official 20k only (our reproduction)	1e-5	96.38%	96.54%
RoBERTa-large + 1M reviews	1e-5	97.24%	97.21%
ELECTRA-large + 1M (long reviews in sliding windows)	1e-5	97.37%	97.22%
DeBERTa-v3-large + 1M	8e-6	97.51%	97.33%
Stacked ensemble: ELECTRA + DeBERTa + 1 MB n-gram	–	97.54% (5-fold CV)	97.40%

- All large models: 1 epoch on 1,020,000 reviews, batch 32, bfloat16, H100; RoBERTa-large about 490 reviews/s, DeBERTa about 218/s.
- Stacker weights (logit scale): DeBERTa 0.577, ELECTRA 0.205, n-gram 0.042.
- 95% bootstrap CI 97.21–97.58%; de-duplicated test (24,479 reviews): 97.40%.
- Exact McNemar vs best single model: 41 vs 24 discordant reviews, p = 0.046. Vs official-data RoBERTa-large: 359 vs 145, $p \approx 5.6 \times 10^{-22}$.
- Reference points: ERNIE-Doc 97.1% (best peer-reviewed), XLNet 96.8%, LlamBERT 96.68% (extra data); our RoBERTa-large reproduction matches the published 96.54% baseline.

- **How to state it:** “97.40% on IMDb test using about 1M extra rating-labelled IMDb reviews, de-duplicated and movie- and author-disjoint from test”: an extra-training-data result, single training runs, and about 2.9% of IMDb test labels are wrong (Northcutt et al. 2021), so ~97% is the practical ceiling.

8. Learning to defer (Phases 4–5)

Deferral labels

Hub, slice	Hub fixes (+1)	Hub breaks (−1)	No change (0)
DistilBERT 20k, v1 slice (device v1)	4.10%	3.90%	92.00%
DistilBERT +1M, v2 slice (device v2)	14,942 (4.98%)	4,564 (1.52%)	280,494
RoBERTa-base +1M, v2 slice (device v2)	16,276 (5.43%)	3,732 (1.24%)	279,992

Router training: 250,000 reviews of the slice; regularisation C from {0.1, 0.3, 1, 3, 10} chosen by accuracy at 10% sent on the other 50,000 (C = 0.3 for the weak hub, 0.1 for the 1M hubs). A variant that sees only the confidence and length bins was also trained.

Evaluation protocol

- For a target send share t, the threshold is the (1 − t) quantile of the router's scores on the 5,000 validation reviews, applied once to the 25,000 test reviews (test send shares land within 2 points of t).
- Also reported: the send share with the best validation accuracy.
- Paired comparison: exact McNemar test on the 25,000 test predictions; 5 operating points × 3 hubs = 15 tests, Bonferroni level 0.05 ÷ 15 = 0.0033.

Results: test accuracy (%), confidence / learned (McNemar p)

Hub (alone)	5% sent	10%	20%	30%	Best rate on val
DistilBERT 20k (93.51)	93.63 / 93.64 (0.94)	94.05 / 93.92 (0.04)	93.91 / 93.98 (0.28)	93.81 / 93.95 (0.10)	94.07 / 93.99 (0.22); 12% / 17% sent
DistilBERT +1M (95.83)	93.71 / 93.68 (0.64)	94.70 / 94.75 (0.24)	95.48 / 95.54 (0.07)	95.72 / 95.72 (1.00)	95.83 / 95.82 (0.57); 77% / 78% sent
RoBERTa-base +1M (96.72)	93.97 / 93.91 (0.37)	95.14 / 95.23 (0.03)	96.12 / 96.16 (0.28)	96.52 / 96.52 (1.00)	96.68 / 96.72 (0.13); 53% / 98% sent

- Largest gap at any tested point: 0.14 points. The only two $p < 0.05$ point in opposite directions and both fail the 0.0033 bar. RoBERTa-base's +0.10 at 10% is partly a send-share effect (10.7% vs 10.5% sent; at exactly 10% the gap is +0.04).
- **Worked example** (RoBERTa-base, confidence, 10%): threshold = 90th percentile of validation scores = 0.255 (send if the device is under 74.5% sure) → 2,625 test reviews sent (10.50%): 832 fixed, 127 broken, 1,522 already right, 144 wrong either way → (23,079 + 832 − 127) ÷ 25,000 = 95.14%.
- **Why no win:** learned deferral can avoid breaks or find fixes. Strong hubs break only 1.2–1.5%, so the first lever is gone; at 10% sent, 58% of what either router sends was already right and learned deferral doesn't lower that. A perfect router reaches 97.32% at 5% sent, so the headroom is real; the device-side features just don't reveal where the hub is right.
- **Confidence cascade** (cascade.py, n-gram device + 1M hubs, τ within 0.5 pts of the hub on val): 95.45% at 20.0% sent (DistilBERT), 96.11% at 20.5% (RoBERTa-base).

- With the weak 20k hub, routing beats both models: 94.05% at 10.7% sent vs hub alone 93.51%. (An earlier “94.07% at 16%” was the maximum of the test curve, i.e. chosen on test, and is not reported.)

9. Energy (Phase 6)

Device energy per review, with s the share of reviews sent: E(s) = V × [I_cpu × t_local + s × (I_tx × t_tx + I_rx × t_wait)]. Idle and sleep power are excluded.

Parameter	Value	Source
V	3.3 V	ESP32-S3 datasheet v2.2
I_cpu	51.2 mA	Datasheet Table 5-9: 240 MHz, one core
I_tx	283 mA	Datasheet Table 5-7: 802.11n MCS7 at 18.5 dBm
I_rx	88 mA	Datasheet Table 5-7: receive
t_local	5 ms	Placeholder, to be measured
t_tx	10 ms	Placeholder: sending ~1.5 KB on an open connection
t_wait	1 s	Placeholder: radio listening while the hub answers

Answering on the device: 3.3 × 0.0512 × 0.005 = 0.8448 mJ. Each send: 3.3 × (0.283 × 0.010 + 0.088 × 1.0) = 299.74 mJ. Sending everything: 300.58 mJ per review. Because the wait dominates, energy is nearly proportional to s.

Operating points chosen on validation, applied once to test (paper Table III)

Hub	Gap closed	Sent (test)	Test accuracy	mJ / review	Less than send-all
DistilBERT 20k	50%	1.2%	92.93%	4.3	69.7×
DistilBERT 20k	75%	2.1%	93.14%	7.2	41.8×
DistilBERT 20k	90%	3.4%	93.31%	11.1	27.0×
DistilBERT +1M	50%	8.3%	94.41%	25.8	11.7×
DistilBERT +1M	75%	15.0%	95.19%	45.9	6.5×
DistilBERT +1M	90%	26.1%	95.68%	79.0	3.8×
RoBERTa-base +1M	50%	8.3%	94.76%	25.8	11.7×
RoBERTa-base +1M	75%	16.9%	95.89%	51.5	5.8×
RoBERTa-base +1M	90%	25.1%	96.36%	76.0	4.0×

- Worked: RoBERTa-base, 75% of the gap. Target on validation 92.56 + 0.75 × 3.90 = 95.485% → needs 16% sent on validation → on test 16.9% sent, 95.89% → 0.8448 + 0.169 × 299.74 = 51.5 mJ → 300.58 ÷ 51.51 = 5.84×.
- Battery (illustrative 2,000 mAh, 3.7 V = 26,640 J): device only 31.5M reviews per charge; close 75% of the gap 517k; send everything 89k.
- **Scope of the old headline:** the 19 Oct claim “14–19× less energy at the transformer's accuracy” holds only for the weak 93.51% hub, which barely beats the device. With the strong hubs, fully matching the hub needs 43% (DistilBERT) or 92% (RoBERTa-base) sent on the test curve.

10. Compute cost

Itemised per run with “modal billing report” (run IDs matched to the log files). All paid from Modal credits; $0 billed. Every run had a hard time cap set by --max-hours. The H100 rate is $3.95 per hour.

Run	Date	Cost
DeBERTa-v3-large + 1M	19 Oct	$9.03
RoBERTa-large + 1M	19 Oct	$3.71
ELECTRA-large + 1M	19 Oct	$3.68
RoBERTa-large, official 20k	19 Oct	$0.43
Long-review predictions (3 runs)	19 Oct	$0.63
Tokenising (3 runs), one run with no matching log, data upload	19 Oct	$0.25
Record attempt total		**$17.71**
RoBERTa-base hub + 1M	25 Oct	$1.64
DistilBERT hub + 1M	25 Oct	$0.91
DistilBERT tokenising	25 Oct	$0.02
Retrained-hub follow-up total		**$2.56**
All runs		**$20.27**

Modal's October summary shows $20.47; $0.20 is not tied to any run. The scripts assume a $30 budget, so about $9.50 of credits remain. Everything else (device models, routers, analysis, paper, slides, demo) ran free on the laptop (Apple M3 Pro, 18 GB).

11. Corrections and lessons

Corrections made

Issue	Fix
“94.07% at 16% sent” was the best point of the test curve (chosen on test)	Withdrawn; only validation-chosen operating points are reported
Round-1 cleaning missed test-movie reviews, test-author reviews and near-duplicates	Round 2 after an adversarial review; splits_v2
31% of the v1 deferral slice was in the retrained hubs' training data	Deferral slice re-drawn from v2; 0 overlap verified
“14–19× less energy” presented as general	Scoped to the weak hub; strong-hub figures from validation-chosen points
Follow-up cost first reported as $2.76 (change in the monthly total)	$2.56 itemised per run; ROADMAP, paper and slides updated
Routing model called “64 KB” (one of its weight tables)	128 KB (two tables) in paper, Fig. 1, slides, scripts, Doc and ROADMAP
Slide claimed “largest gap at any send rate”	“at the tested send rates” (with the weak hub the curves differ by 0.28 at 60%)
Deck charts used hard-coded “TF-IDF” and “DistilBERT” labels for other models	cascade.py labels outputs by model prefix

Still to correct

- **“23,818 copies of validation or test reviews”** (paper, slides, scripts, ROADMAP): 23,818 is test copies only (IMDb 23,636 + Amazon 182). Validation copies (4,739) and training copies (18,892) were removed as well.
- **ROADMAP Phase 6 row** quotes test-curve figures (“13–14% sent closes 75% of the gap at 40–43 mJ, ~7×”); the paper and slides use the stricter validation-chosen figures (15–17% sent, 5.8–6.5×). Align the ROADMAP to the paper.
- **slides/project_update.pptx** (19 Oct) still shows “14–19×”, “~5% sent matches the hub” (true only for the weak hub), “23,636 copies” and “$17.99 of $30” (itemised: $17.71). Treat it as a historical snapshot.
- **README.md** still describes the original TF-IDF + DistilBERT cascade at the top and doesn't mention the paper, talk or demo.

Engineering lessons

- Public scrapes contain the benchmark: always de-duplicate against val and test, and check movies and authors, not just text.
- DeBERTa loaded in float16 turned every weight into NaN: force float32 when loading, then use bf16 autocast.
- Apple GPUs recompile per input shape: pad batches to multiples of 64.
- Stream large files (23M reviews) and keep token ids as flat int32 arrays to fit in 18 GB of RAM.
- Cache tokens on cheap CPU machines; cap every GPU run's time; checkpoint and resume.
- Choose everything on validation data; report confidence intervals and paired tests, and correct for multiple comparisons.

12. Open items and next steps

1. **Commit the work to git.** The repository has no commits. .gitignore already excludes data/, outputs/models/, .npy files and node_modules; commit the code, paper, slides, demo, ROADMAP and result JSONs.
2. **Make the slide builders self-contained.** slides/build_paper_deck.js and build_scripts_doc.js need a theme helper (apply_theme.js) and the npm packages docx, react, react-dom, react-icons and sharp, which currently live outside the repo. Copy apply_theme.js into slides/ and add the packages to package.json.
3. **Phase 7 hardware:** ESP32-S3 firmware (tokenise, hash, int8 lookup, route), Pi Zero 2 W hub serving an int8 ONNX model over HTTP, INA219 current measurements. Replace the three placeholder timings; measure the hub's latency and int8 model sizes; decide DistilBERT vs RoBERTa-base on measured latency.
4. **Pending text fixes** from section 11 (23,818 wording, ROADMAP Phase 6, README).
5. **Paper details:** author emails or ORCIDs (the last author line currently shows registration numbers), acknowledgment wording, and the target conference's own requirements (page limit, copyright line in the footer).
6. **Talk:** a timed rehearsal (the scripts assume about 130 words a minute, 24.5 minutes total); optionally a “live demo” slide in Part 4.
7. **Research follow-ups:** richer router signals (e.g. disagreement between two small device models), repeated seeds for the record models, evaluation on corrected IMDb test labels, a second dataset with short messages.

13. Deliverables, files and how to rebuild

Deliverables

What	Where	Rebuild
IEEE paper (5 pp, A4 template)	paper/energy_aware_routing_ieee.docx and .pdf	paper/make_figures.py, then paper/build_paper.py <template.docx>
Paper figures and energy table	paper/figures/, paper/energy_table.json	paper/make_figures.py
Presentation, 22 slides, 3 presenters	slides/paper_presentation.pptx and .pdf	slides/build_paper_deck.js (writes presenter_scripts.md too)
Presenter scripts	slides/presenter_scripts.docx, .pdf, .md	slides/build_scripts_doc.js (after the deck)
Presenter scripts, shared Doc	https://claude.ai/artifact/HRY4ZgeDs3Yy4qtbqyAAeA	Edited online; overview, scripts, Q&A
Laptop demo	demo/ (app.py, engine.py, index.html, README.md)	.venv/bin/python -I demo/app.py → http://localhost:8770
Roadmap and results narrative	ROADMAP.md, README.md	Edited by hand
Earlier decks	review2_update.pptx, slides/project_update.pptx	build_deck.js, slides/build_project_update.js

Demo

The demo runs the real models on the laptop: the v2 256 KB device model, the RoBERTa-base router and the RoBERTa-base +1M hub (weights downloaded to outputs/models/roberta_base_1m_best, 499 MB). For each review it shows the device's answer and the words that drove it, whether it is sent and why, the hub's answer, and the simulated energy (processing 0.84 mJ; radio transmit 9.34 mJ and receive 290.4 mJ when sent), plus session totals, reviews per charge, and a 100-test-review batch run. demo/engine.py self-checks against the paper: device predictions match exactly; confidence routing at 10% gives 95.136% (paper 95.136%); learned 95.232% (paper 95.228%, one review apart). demo/README.md has a two-minute talk track.

Key source files

File	Role
src/prepare_data.py, src/fast_model.py, src/train_transformer.py, src/cascade.py	Phase 0: splits, TF-IDF + LR, DistilBERT, cascade (cascade.py now takes --fast-prefix/--bert-prefix/--tag)
src/build_pool.py, src/maas_movies.py, src/pool_splits.py, src/pool_hygiene.py	Phase 1: pool, test-movie list, v1 splits, round-2 cleaning and splits_v2
src/esp32_feasibility.py, src/chip_model.py, src/ngram_v2.py	Phase 2: size study, device model v1, v2 models (also score the v2 deferral slice and save weights)
src/train_hub.py, src/modal_app.py, src/predict_hub.py	Hub and record models on Modal; long-review sliding windows
src/hub_labels.py, src/defer_router.py	Phases 4–5: hub labels; routers, operating points, McNemar (--chip-prefix/--hub-prefix/--defer-cache/--tag)
src/energy_curves.py	Phase 6: energy model (--tag)
src/ensemble.py	Record ensemble, bootstrap CI, McNemar
src/dashboard.py	Live job dashboard (localhost:8765)
outputs/*.json, outputs/logs/	Every result and run log quoted in this handoff
papers/ + papers/references.bib	Every cited source, grouped by purpose, with BibTeX

Reproduce the retrained-hub results

```
python -m modal run --detach src/modal_app.py::train --gpu H100 --max-hours 1 \
  --args "--model roberta-base --pool-reviews 1000000 --prefix roberta_base_1m --lr 2e-5 --
batch-size 32 --accum 1"
python -m modal volume get ai-sentiment outputs/roberta_base_1m_<split>_proba.npy outputs/
python -I src/ngram_v2.py
python -I src/defer_router.py --chip-prefix ngram256KB --defer-cache defer_v2 --hub-prefix
roberta_base_1m --tag roberta_base_1m
python -I src/energy_curves.py --tag roberta_base_1m
python src/cascade.py --fast-prefix ngram256KB --bert-prefix roberta_base_1m --benchmark 0 -
-tag roberta_base_1m
```

Earlier steps (data download, pool, device model, record attempt) are listed step by step in README.md (steps 1–20). Storage: data/ is 30 GB (16 GB raw downloads, 8.7 GB pool, 5.4 GB caches); outputs/ is 0.8 GB. Modal volume “ai-sentiment” holds the uploaded splits, token caches and every cloud checkpoint (outputs/models/*).

Where else context lives

- ROADMAP.md: phase table, Phase 5 follow-up section with the full comparison table.
- Session hand-over note: ~/Library/Application Support/Claude/side-session-notes/local_fdf17b9b-…/phase5_followup_retrained_hub.md.
- papers/README.md: why each cited paper is in the project.